

Mohamad Abadi

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EDUCATION

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|---|-----------------------|
| University College London (UCL) – London, United Kingdom | Sep. 2024 – Sep. 2025 |
| <ul style="list-style-type: none">MSc in Chemical Process Engineering (Distinction)Thesis: <i>Modelling Tyre Pyrolysis in a Rotary Kiln using Induction-Heated Susceptors</i>Modules: Process Engineering Modelling and Design, Advanced Process Optimisation, Energy Systems | |
| American University of Beirut (AUB) – Beirut, Lebanon | Sep. 2019 – May. 2024 |
| <ul style="list-style-type: none">BEng in Chemical Engineering with a minor in Chemistry (Graduating GPA: 3.8/4.0)Placed on Dean's Honor List in the last four semesters | |

PROFESSIONAL EXPERIENCE

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| Operations Manager – BNS Real Estate LTD – London, United Kingdom | Aug. 2024 – Present |
| <ul style="list-style-type: none">Manage end-to-end operations across 30+ tenanted residential properties, coordinating tenants, landlords, contractors, building management, and a small internal teamReduced contractor spend by 40% by benchmarking and replacing the incumbent maintenance provider, and cut monthly storage costs by 73% through Excel-based space optimisationManaged property projects and incidents end to end, including renovation planning, AutoCAD layouts, insurance claims, contractor coordination and temporary tenant relocation | |
| Junior Consultant – FREE Energy – Remote | Sep. 2023 – Jan. 2025 |
| <ul style="list-style-type: none">Delivered energy audits to ISO 50002, EN 16247 and ASHRAE 211 across 20+ public and private sector sites in the UAE, Saudi Arabia and Lebanon, spanning hospitality, government, education, commercial and industrial facilitiesBuilt consumption baselines from utility bills and distribution board logger data, including deriving cost of energy from generator fuel curves at off-grid sites with no tariffDeveloped and costed energy conservation measures across HVAC, chillers, lighting, pumps and BMS, priced through bills of quantities and appraised on NPV, IRR and payback alongside avoided emissions in tCO₂e, identifying up to 579,000 kWh and AED 116,000 in annual savings at a single siteDesigned and delivered energy efficiency training in English and Arabic, including a train-the-trainer programme for facilities staff and a seven-competency building insulation course with trainer manuals and assessments | |
| Lab Research Intern – University of Barcelona – Barcelona, Spain | Jun. 2023 – Aug. 2023 |
| <ul style="list-style-type: none">Conducted 24 systematic experiments involving the preparation, processing, and testing of polymer-based materialsUsed FTIR and viscosity analysis to troubleshoot failed reactions and optimise catalyst concentration, achieving successful CAP synthesis after initial failed trialsProduced biodegradable cellulose acetate propionate (CAP) films as a sustainable alternative to petroleum-based packaging materials | |

EXTRACURRICULAR ACTIVITIES & LEADERSHIP EXPERIENCE

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| Senior Editor & Social Media Representative – AUB Outlook – Beirut, Lebanon | Sep. 2020 – May. 2024 |
| <ul style="list-style-type: none">Progressed from writer to Senior Editor, managing a team of 8+ writers and editors across sports and technology coverageAuthored 10+ published articles, covering university sports, technology trends, and student lifeLaunched and filmed a series interviewing sports team captains, increasing university sports engagement and visibility | |
| Class Representative – American Institute of Chemical Engineers – Beirut, Lebanon | Sep. 2022 – Sep. 2023 |
| <ul style="list-style-type: none">Planned a 5-day industrial trip for 40 students to visit 3 chemical engineering plants in JordanCommunicated classmates' concerns and requests to the faculty chairperson, resulting in curriculum changesDirected and starred in 3 promotional videos for AIChE's social media campaign, increasing engagement by 80% | |

PROJECTS

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| Master's Thesis – Modelling Tyre Waste Pyrolysis in a Rotary Kiln | Nov. 2024 – Sep. 2025 |
| <ul style="list-style-type: none">Developed two 1-D MATLAB kiln models comparing conventional and induction-assisted heating, reducing modelled specific energy demand from 1,400 to 1,270 kJ/kg while identifying a trade-off as conversion decreased from 100% to 88.5% | |
| Molecular Fingerprint Clustering | Feb. 2025 – Mar. 2025 |
| <ul style="list-style-type: none">Analysed 20,609 molecules using 2,048-bit Morgan fingerprints, K-Means, PCA and RDKit in Python, identifying three chemically interpretable groups and evaluating cluster selection using inertia and silhouette scores | |
| Renewable Energy Platform and Anaerobic Digestion Plant in Lebanon | Sep. 2023 – May. 2024 |
| <ul style="list-style-type: none">Co-designed an integrated solar, wind, wave and biogas system in a team of five, sizing an anaerobic digestion plant for 2,000 tonnes of organic waste a day across 27 digesters and optimising 27.6 MW of generation, 241.8 GWh a year | |

SKILLS & INTERESTS

- Technical Skills:** Excel, Microsoft Office, Python, Aspen HYSYS & Plus, MySQL, MATLAB, GAMS, gPROMS, AutoCAD, HelioScope
- Languages:** Arabic (Native), English (Fluent)